

Self-Reinforcing Gradient Antifouling Coating with High Adhesion, On-Demand Peelability, and Superior Recyclability

Zhenqiang Zhang, Bin Zhang, Zhipeng Zhang, Qingyi Xie, Guoliang Zhang, Guangzhao Zhang, and Chunfeng Ma*

On-demand peelable antifouling coatings can address the issue of coating failure due to high biofouling pressure through simple removal and meet the temporary antifouling needs of underwater equipment. However, such materials are rare due to the challenge of simultaneously fulfilling the requirements for high adhesion, on-demand peelability, recyclability, and surface antifouling functionality. Herein, a novel strategy is presented for preparing a gradient antifouling coating with high adhesion, on-demand peelability, and superior recyclability that involves the modular construction of three components that work together synergistically. Self-reinforcing thiol-capped polysilsesquioxane (POSS-SH, module 1) reacts with isocyanate-terminated polyurethane (NCO-PU, module 2) to form a cross-linked network with strong cohesion, and reactive amphiphilic polymer (RAP, module 3) self-enriches on the coating surface to exert fouling resistance against proteins, bacteria, diatoms, and biofilms. Meanwhile, the coating achieves strong adhesion to a substrate by hydrogen bonds and high recyclability owing to dynamic thiourethane bonds. By leveraging the synergies between modules, the gradient coatings showcase superior antifouling ability, high adhesion strength ($\approx 3.5\text{--}9.0\text{ MPa}$), easy peelability ($\approx 1.5\text{--}5.5\text{ N}\cdot\text{cm}^{-1}$), high transparency ($>85\%$ transmittance), and recyclability. It is expected to be useful as a temporary or durable antifouling coating in marine applications such as underwater sensors and robots.

various effective strategies such as self-polishing coatings,^[7,8] fouling release coatings,^[9,10] fouling resistant coatings,^[11,12] dynamic surfaces,^[13] and biomimetic surfaces.^[14–16] In general, an antifouling coating must meet the fundamental requirements of both good antifouling performance and strong adhesion to the substrate that can resist harsh environments.^[17] Nevertheless, high biofouling pressure may cause the coating to fail and form persistent biofouling that is difficult to remove.^[18,19] In such cases, directly peeling the coating from the substrate in one continuous sheet is a simple and effective solution. However, the above coatings lack peelability and can only be removed by complex and time-consuming physical methods (e.g., mechanical sanding and sandblasting) or chemical methods (e.g., acid, alkali, and solvent treatment).^[20,21] Additionally, coatings with on-demand peelability would be able to fulfill the demand for not only durable antifouling protection but also temporary or short-term antifouling protection of a few days to several months. Their easy removal would simplify

subsequent functionalization of the device surface. Therefore, developing antifouling coatings with high adhesion and on-demand peelability would have substantial economic and environmental benefits.

Peelable coatings that can actually be peeled off the substrate in a continuous sheet are widely used for temporary surface protection in aerospace and automotive industries, military equipment, and precision instruments.^[22] To achieve peelability, the adhesion strength of a coating to a substrate must be weaker than the force required to peel it off.^[23] The main approach to developing peelable coatings is to mix a polymer resin system with additives such as fatty acid esters, fatty acid salts, waxes, surfactants, and organosilicon compounds to reduce adhesion between the coating and the substrate.^[24] However, these additives are unstable within the resin and can even contaminate the protected substrate surface. More importantly, for marine applications, since the coating will be exposed to seawater washout over a long period of time, a peelable antifouling coating should maintain good

1. Introduction

Biofouling is the attachment and accumulation of organisms to equipment threatening their normal operation, and it is of great concern in fields such as biomedical engineering, biosensors, and maritime industries owing to its potential to cause huge economic losses.^[1–4] The most common approach to preventing biofouling is to use antifouling coatings,^[5,6] which can employ

Z. Zhang, B. Zhang, Z. Zhang, Q. Xie, G. Zhang, G. Zhang, C. Ma
Faculty of Materials Science and Engineering
South China University of Technology
Guangzhou 510640, P. R. China
E-mail: msmcf@scut.edu.cn

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adfm.202508400>

DOI: 10.1002/adfm.202508400

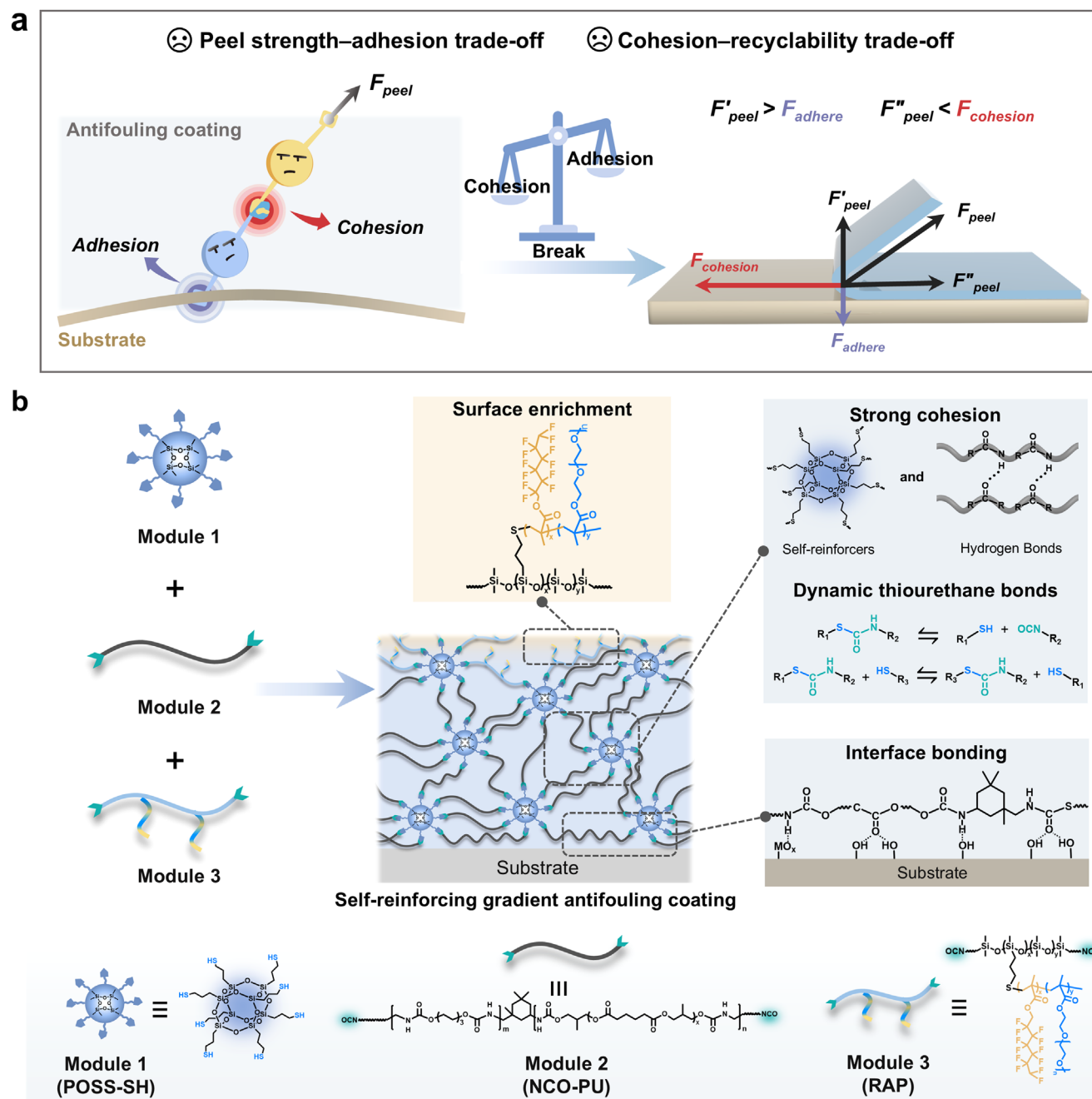


Figure 1. a) Major challenges and design principles of a peelable antifouling coating. b) Schematic description of the preparation process of self-reinforcing gradient antifouling coatings with high adhesion, on-demand peelability, and superior recyclability.

adhesion to the substrate during its lifespan.^[25,26] Therefore, new strategies are needed to balance adhesion and peelability.

Actually, the adhesion and peelability are mutually contradictory. To achieve a coating with high adhesion while maintaining peelability, controlling the cohesion within the material is critical. **Figure 1a** shows an analysis of the forces involved in the process of peeling a coating from a substrate. The peel force (F_{peel}) can be decomposed into vertical (F'_{peel}) and horizontal (F''_{peel}) components. To peel the coating from the substrate, F'_{peel} must overcome the downward adhesion force (F_{adhere}) between the sub-

strate and the coating. If F_{adhere} greatly exceeds the F'_{peel} , the peeling process will not be successful. However, if F_{adhere} is too low, the coating may detach spontaneously. Therefore, a suitable F_{adhere} is required to enable the coating to be peeled off while simultaneously fulfilling the adhesion requirements of the application scenario. Meanwhile, F''_{peel} requires the coating to have sufficient tensile strength and toughness to prevent tearing or breaking. Therefore, the coating needs to possess strong cohesion to ensure it does not easily rupture under lateral stress. Introducing chemical covalent cross-linking into the linear network of

a material is one of the most effective methods to enhance the cohesion of the coating.^[27,28] However, covalent cross-linking is irreversible, which limits the recyclability of the material and results in its disposal at the end of its lifespan, which wastes resources and increases the environmental impact. Thus, another challenge for the design of peelable antifouling coatings is balancing the trade-off between cohesion and recyclability.

Stereoscopic polysiloxanes with 3D structures such as branched,^[29] cage-like,^[30,31] ladder-like,^[32,33] or spherical^[34] are predominantly synthesized by sol-gel chemistry and feature rigid inorganic cores surrounded by organic components, which can be functionalized or cross-linked via different polymerization methods, such as free radical, cationic, epoxy-amine, and thiol-ene reactions.^[35] In particular, cage-like polysilsesquioxane (POSS) forms a distinct 3D network characterized by a rigid Si–O–Si core encircled by various organic functional groups.^[36,37] POSS can be used to notably improve the cohesion of polymeric materials by extensive cross-linking as well as providing a reinforcing effect owing to its rigid structure.

Herein, we designed a novel self-reinforcing gradient antifouling coating with high adhesion, on-demand peelability, and superior recyclability. As shown in Figure 1b, the basic building blocks are thiol-functionalized POSS (POSS-SH, module 1) and isocyanate-terminated polyurethane (NCO-PU, module 2). POSS-SH is a reversible self-reinforcer that can form dynamic covalent bonds with NCO-PU, and its Si–O–Si inorganic core strengthens the cohesion of the material, enabling the coating to be peeled off. The urethane and ester bonds in module 2 further improve the cohesion and form strong hydrogen bonds with the substrate to provide high adhesion. Furthermore, we introduced an antifouling functional module 3—reactive amphiphilic polymer (RAP)—to impart antifouling properties to the coating surface. The inherent low surface energy of RAP facilitates spontaneous self-enrichment driven by thermodynamic and kinetic driving forces. Concurrently, its terminal isocyanate can cross-link with the polymer network to improve fouling resistance while maintaining high adhesion, on-demand peelability, and recyclability. These unique combinatory properties enable the coating not only to address the issue of coating failure caused by high biofouling pressure through simple removal but also to be applied for the temporary antifouling of underwater equipment, which is not the case with the aforementioned antifouling coatings. Simulations and experiments were performed to evaluate the influence of the network structure on the peelability and adhesion of the coating, the degree to which POSS enhances the cohesion, and the synergistic effects between modules. Our aim is to propose a novel strategy to prepare gradient antifouling coatings with excellent substrate adhesion, peelability, and recyclability.

2. Results and Discussion

2.1. Design and Preparation of Gradient Coatings with High Adhesion and On-Demand Peelability

Simultaneous realization of high adhesion and peelable properties in a coating requires to rationally choose and design the components of the polymer network. The structure of module 1 (POSS-SH) was characterized by ¹H and ²⁹Si nuclear magnetic

resonance (NMR) and dynamic light scattering (DLS). The ¹H NMR spectrum (Figure S1, Supporting Information) had peaks at 0.76, 1.71, and 2.56 ppm, which were assigned to the three kinds of methylene on POSS-SH, and a peak at 1.40 ppm attributed to the –SH group. The ²⁹Si NMR spectrum (Figure S2, Supporting Information) showed that the terminal unit (*T*₁) had disappeared and the dendritic unit (*T*₃) was much higher than the linear units (*T*₂), which indicated a high degree of condensation. The high degree of condensation implies that minimal unstable Si–OH groups remain for side reactions during thiol–isocyanate reaction. The DLS results (Figure S3, Supporting Information) showed that the POSS-SH had an average size of ≈2.5 nm, which was beneficial to the exchange of dynamic covalent bonds. These results confirm the successful synthesis of POSS-SH. Module 2 (NCO-PU) containing polyester soft segments had a number-average molecular weight of 5500 g/mol (Figure S4, Table S1, Supporting Information). Fourier-transform infrared (FTIR) spectra confirmed the successful synthesis of NCO-PU (Figure S5, Supporting Information). The combination of module 1 and module 2 allows the preparation of coatings with high adhesion and on-demand peelability (SiPU). The structure of SiPU was confirmed by FTIR spectra (Figure S5, Supporting Information), revealing absorption peaks at 3360 cm^{−1} attributed to N–H bonds, 1700 cm^{−1} corresponding to C=O bonds, and 1120 cm^{−1} indicative of Si–O–Si asymmetric stretching. Notably, the absence of the –NCO peak at 2270 cm^{−1} in the spectrum suggested the complete consumption of NCO-PU. Thus, these spectral analyses confirmed the successful synthesis of SiPU.

To elucidate the contributions of individual modules to the mechanical properties of the developed coating, two additional coatings were designed (Figure 2a). SiPU-PPG was prepared by substituting poly(2-methyl-1,3-propylene adipate) diol (PA2000) in SiPU with polypropylene glycol (PPG2000). Linear polyurethane (LPU) was prepared by substituting POSS-SH in SiPU with 1,6-hexanediol (HDO) (Figures S6,S7, Table S2, Supporting Information). The number-average molecular weights of NCO-PU-PPG precursor and LPU were 5000 and 22100 g/mol, respectively. FTIR spectroscopy confirmed the successful synthesis of SiPU-PPG and LPU (Figures S8,S9, Supporting Information).

Figure 2b compares the mechanical properties of LPU, SiPU-PPG, and SiPU obtained through experimental tests. LPU exhibited the highest elongation at break but had a low tensile strength of 3.0 MPa. In contrast, SiPU and SiPU-PPG had tensile strengths of 21.4 and 10.0 MPa, respectively, which indicated that self-reinforcing POSS-SH greatly enhanced the material strength (Figure 2c). Specifically, SiPU, prepared by combining a polyester-based polyurethane prepolymer (NCO-PU) with POSS-SH, showed significantly superior tensile strength and toughness compared to LPU and SiPU-PPG, demonstrating strong cohesion (Figure 2d). Next, pull-off adhesion tests were performed on the coatings, and LPU, SiPU-PPG, and SiPU exhibited adhesion strengths of 6.8 ± 1.4, 9.6 ± 1.2, and 6.9 ± 1.1 MPa, respectively, on glass (Figure 2e). All coatings exhibited high adhesion strengths of >6 MPa, which was attributed to the formation of strong hydrogen bond interactions between the polar bonds in the polymer network (e.g., urethane, thiourethane, or ester) and the polar groups on the glass surface. When the coatings were peeled off the glass, the LPU coating underwent excessive stretching and deformation, the SiPU-PPG coating cracked,

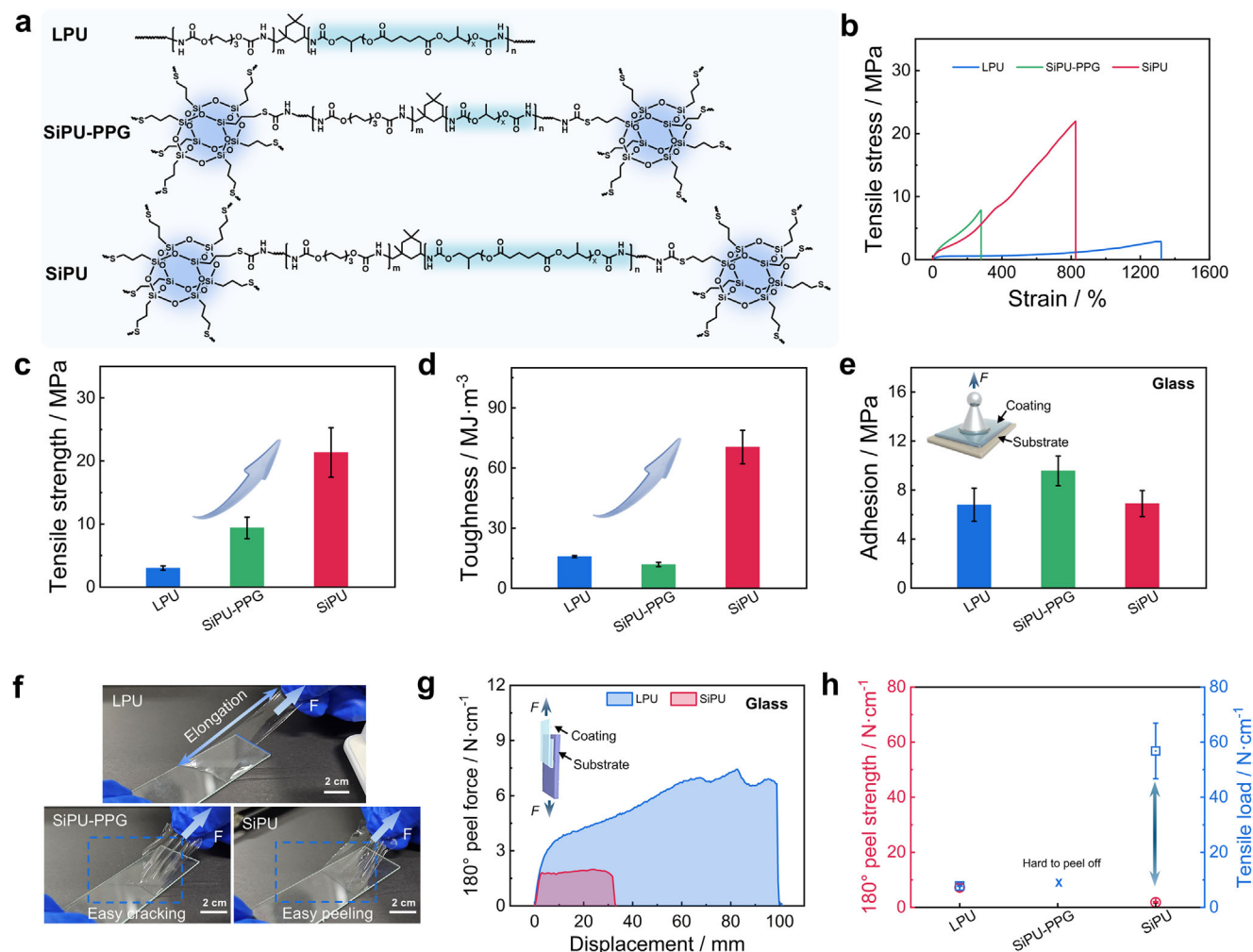


Figure 2. a) Chemical structures of LPU, SiPU-PPG, and SiPU. b) Stress–strain curves of LPU, SiPU-PPG, and SiPU and the corresponding c) tensile strength and d) toughness. e) Adhesion strength of LPU, SiPU-PPG, and SiPU coatings on the glass substrate. f) The peeling process of the LPU, SiPU-PPG, and SiPU coatings. g) Peeling curves of LPU and SiPU coatings on the glass substrate. h) 180° peel strength and tensile load of LPU and SiPU coatings.

and the SiPU coating was easily peeled off in a continuous sheet (Figure 2f). Further analysis was conducted through a 180° peeling test. A 180° peeling test showed that LPU had a much higher peel strength than SiPU and a much smaller difference between the peel strength and tensile load, which indicates that POSS-SH helped enhance the peelability of SiPU (Figure 2g,h). Note that the pull-off adhesion strength of LPU and SiPU on glass appears similar, but in the 180° peel test, the peel strength of LPU is higher than that of SiPU. The possible explanation for this discrepancy is that the pull-off adhesion test measured the detaching behavior of the coatings after being subjected to vertical directional forces, whereas the 180° peeling test examined the detaching behavior of the coatings after being subjected to parallel directional forces. Therefore, the pull-off adhesion is mainly determined by the interface interaction, but the 180° peeling strength is determined by both interface interaction and cohesion energy. During the peeling process, the weak cohesion of LPU caused it to large-scale deformation during the peeling process, necessitating more peel energy for removal and resulting in higher peel

strength than that of SiPU. In contrast, the hydrogen bonding interactions in the SiPU network ensured high adhesion, while the introduction of POSS enhanced cohesion and made the coating much easier to peel. Additionally, SiPU-PPG also had a higher adhesion, but the insufficient cohesion and poor toughness caused it to crack when peeled.

Molecular dynamics (MD) was used to obtain microscopic insights into the adhesion and peelability of SiPU. As shown in Figure 3a,b, the cohesive energy density (CED) of SiPU was 8.47 GJ·m⁻³ at room temperature compared with only 4.38 GJ·m⁻³ for LPU, which demonstrates its strong cohesion (see also Figures S10,S11, Supporting Information). This was attributed to the abundant cross-linking sites of POSS, which afforded strong dynamic covalent bonds as well as dense H-bonding interaction with the PU. SiPU also had a higher CED than SiPU-PPG at 7.89 GJ·m⁻³, which was attributed to the enhancement of the ester bond in NCO-PU (module 2). Figure 3c compares the average H-bonding numbers of the bulk phase and shows that SiPU had the highest H-bonding number followed by SiPU-PPG

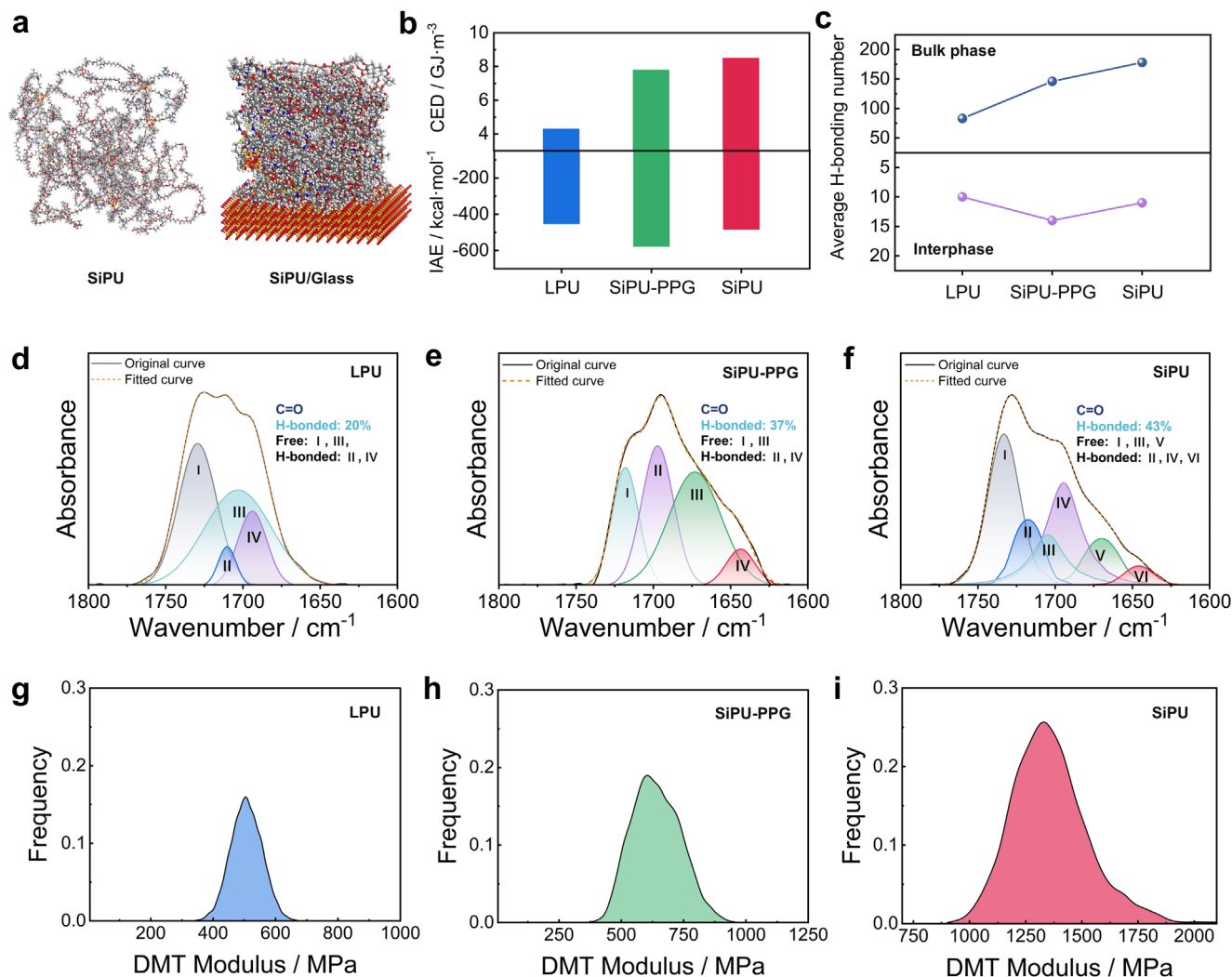


Figure 3. a) Optimized stable conformations with a minimum energy of SiPU and SiPU/Glass. b) CED, IAE, and c) average H-bonding number of LPU, SiPU-PPG, and SiPU. FTIR spectra in the C=O stretching vibration region of d) LPU, e) SiPU-PPG, and f) SiPU. DMT modulus statistic curves of g) LPU, h) SiPU-PPG, and i) SiPU.

and LPU, which further supports the above results. The interfacial adhesion energy (IAE) was also calculated by MD simulation using a glass surface as a model substrate. The change in IAE was different from the CED: SiPU-PPG > SiPU ≈ LPU, indicating stronger interactions between SiPU-PPG and the glass substrate. The average H-bonding numbers of the interphase are consistent with the trend of the IAE. Although SiPU and LPU had lower IAEs than SiPU-PPG, they were still relatively high at >450 kcal·mol⁻¹, which indicates that they still provide a high adhesion strength. Notably, both LPU and SiPU exhibited similar IAE and average H-bonding numbers, which accounts for their comparable high adhesion strength in the pull-off adhesion test. Meanwhile, SiPU demonstrated significantly higher CED than LPU, which explains why SiPU is more easily peeled than LPU. To further explain the contribution of the self-reinforcing POSS-SH to the polymer network, we analyzed the H-bonds in the network by FTIR spectra of the carbonyl (C=O) region. The C=O stretching region was deconvoluted into multiple subpeaks

in the wavenumber range of 1780–1600 cm⁻¹. The C=O absorption band of LPU and SiPU-PPG were deconvoluted into four subpeaks while that of SiPU was deconvoluted into six subpeaks (Figure 3d–f). The subpeaks at 1730, 1705, and 1670 cm⁻¹ were assigned to free C=O groups while the subpeaks at 1715, 1695, and 1645 cm⁻¹ were assigned to H-bonded C=O in the carbonate group, urethane group, and thiourethane group, respectively. Quantitative analysis indicated that the proportion of H-bonded C=O reached 43% in SiPU compared with only 20% in LPU and 37% in SiPU-PPG. This indicates that the introduction of self-reinforcing POSS-SH induced the formation of H-bonds, and the combination of POSS-SH (module 1) with NCO-PU (module 2) formed denser H-bonds, which further improved the strength and toughness of the coating.

Atomic force microscopy (AFM) nanomechanical mapping was employed to compare the Derjaguin–Muller–Toporov (DMT) modulus images of the coatings. LPU had a relatively homogeneous network structure whereas SiPU-PPG and SiPU exhibited

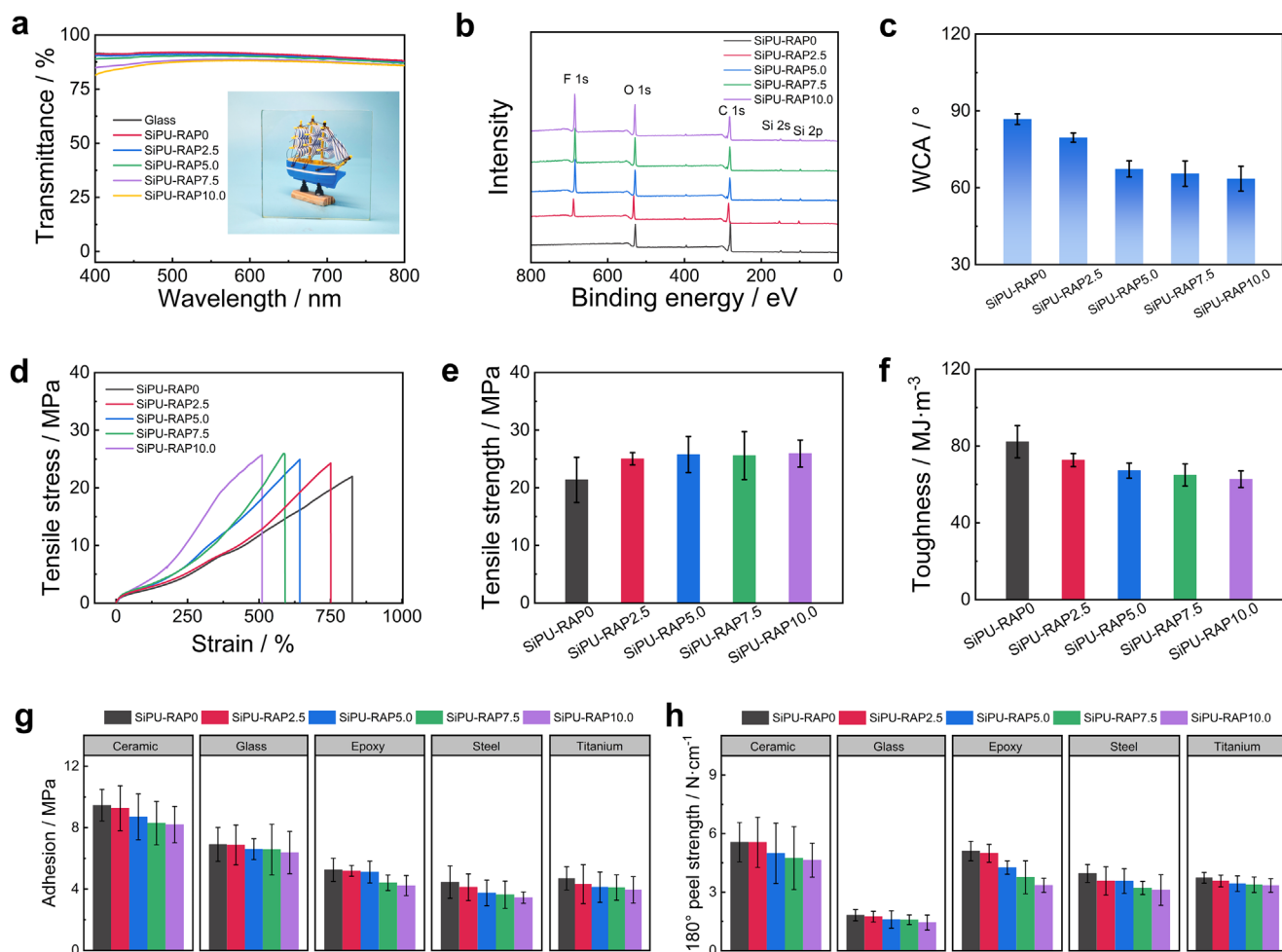


Figure 4. a) UV-vis transmission spectra of glass and SiPU-RAP x (inset photograph showing the transparent SiPU-RAP10.0). b) XPS scan spectra of SiPU-RAP x coatings. c) WCAs of SiPU-RAP x coatings. d) Stress-strain curves of SiPU-RAP x and the corresponding e) tensile strength and f) toughness. g) Adhesion strength and h) 180° peel strength of SiPU-RAP x coatings on various substrates.

nanoscale phase separation with irregular regions corresponding to low (dark) and high (bright) moduli (Figure S12, Supporting Information). SiPU had a wide modulus distribution of 870–1900 MPa, which can be attributed to the large modulus difference between rigid POSS-SH and soft phase. LPU had a narrow modulus distribution of 350–650 MPa, which indicates that self-reinforcing POSS-SH enhanced the cohesive energy and thus the modulus and degree of phase separation in SiPU. Note that the modulus distribution of SiPU was also larger than that of the SiPU-PPG (380–950 MPa), indicating that the synergistic effect between modules further improves the cohesion energy of the materials (Figure 3g–i). The above results indicate that the modular construction strategy and utilization of self-reinforcing POSS helped SiPU balance the trade-off between peelability and adhesion.

3. The Introduction of RAP for Antifouling Ability

Based on the modular construction and self-enrichment strategy, we can easily endow SiPU with programmable functionality by integrating functional modules into the network. For example, by

employing RAP to replace a portion of the PU precursor in SiPU, SiPU endowed with antifouling ability can be achieved. The low-surface-energy functional moieties (fluorocarbon and polydimethylsiloxane) of isocyanate-terminated RAP can self-enrich on the SiPU surface during the solvent evaporation, and the amphiphilic telomer side chains exert biofouling resistance (Figure S13, Supporting Information). Different amounts of RAP were added to SiPU, and the resulting coatings were designated as SiPU-RAP x . The successful synthesis of RAP and SiPU-RAP x were verified through ^1H NMR and FTIR spectroscopy (Figures S14–S17, Supporting Information). To assess the integrity of cross-linking networks, the gel content of SiPU-RAP x was measured. All samples exhibited only swelling and minimal dissolution in THF solvent, which resulted in high gel contents of >95% (Figure S18, Supporting Information). Figure 4a shows the transmittance spectra of SiPU-RAP x coatings applied to glass slides. All SiPU-RAP x coatings demonstrated comparable transmittance values exceeding 85% at a wavelength of 500 nm, which was attributed to the size of nanoclusters and the RAP phase domain being smaller than the wavelengths of visible light. The inset showed the typical sample SiPU-RAP10.0 placed in front of

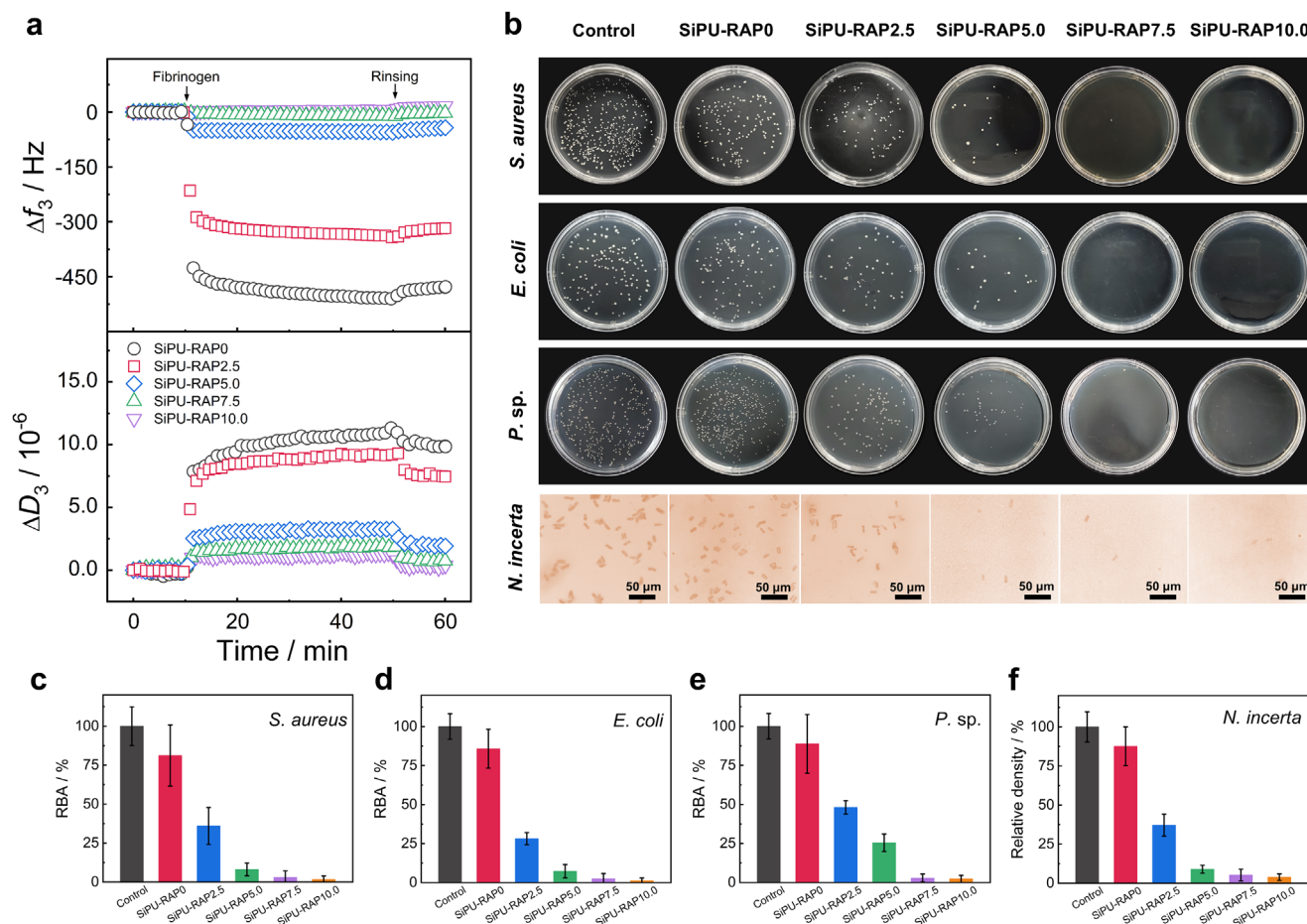


Figure 5. a) QCM-D curves of SiPU-RAP_x coatings. b) Agar plate photographs of bacterial colonies (*S. aureus*, *E. coli*, and *P. sp.*) and microscopic images of the diatom (*N. incerta*) after incubation with SiPU-RAP_x coatings. c–e) The corresponding relative bacterial adhesion (RBA) of *S. aureus*, *E. coli*, and *P. sp.* f) Relative density of *N. incerta*.

the model ship. The optical clarity was not reduced, indicating the high transparency of the coating. Notably, the transmittance of SiPU-RAP10.0 showed a minimal decline even after immersion in artificial seawater for 3 days (Figure S19, Supporting Information).

X-ray photoelectron spectroscopy (XPS) was used to investigate the surface composition of the coatings (Figure 4b). The peak at 685.7 eV corresponds to F (1s) from the fluorocarbon groups. No F (1s) signal was observed for SiPU-RAP0 because it lacked fluorocarbon content, and the intensity of the F (1s) peak increased with the RAP content. Notably, all coatings except for the SiPU-RAP0 showed higher surface F values than the bulk (Table S3, Supporting Information), which suggests that the low surface energy of the fluorocarbon drove the RAP to migrate to the coating surface and resulted in a gradient antifouling coating. To further confirm RAP loading in SiPU, the surface water contact angles (WCAs) of SiPU-RAP_x coatings were measured. As shown in Figure 4c, the WCAs decreased slightly with the incorporation of RAP, which indicates that more PEG chains were driven to the coating surface by linkage with the fluorocarbon.^[38]

The mechanical performance of SiPU-RAP_x was examined comprehensively. Figure 4d shows that the elongation at break

decreased with increasing RAP content. This is likely because some PU segments were replaced by RAP, which has lower molecular weight and multiple side chains that hinder the movement of the soft PU segments.^[39,40] Figure 4e shows that the tensile strength of SiPU-RAP_x slightly increased with increasing RAP content from ≈ 21.4 to ≈ 25.9 MPa, which can be attributed to the shorter siloxane backbone of RAP and its urea bonds that could form strong hydrogen bonds. Correspondingly, the SiPU-RAP_x maintained a high mechanical toughness with increasing RAP content (Figure 4f). In particular, SiPU-RAP10.0 (0.5 g) could withstand a weight of 5 kg for more than 10 min without breaking (Figure S20, Supporting Information).

Furthermore, enrichment of the RAP on the coating surface can lead to good antifouling performance without compromising the adhesion and peelable properties of the coating. Figure 4g shows that the adhesion to various substrates decreased slightly as the RAP content was increased, but SiPU-RAP10.0 still exhibited strong adhesion to ceramic (≈ 8.2 MPa), glass (≈ 6.4 MPa), epoxy (≈ 4.2 MPa), steel (≈ 3.4 MPa), and titanium (≈ 3.9 MPa). This can be attributed to the strong hydrogen bonding interactions between the SiPU-RAP_x coating and the polar substrate surfaces, and substrates with a higher polarity resulted in stronger

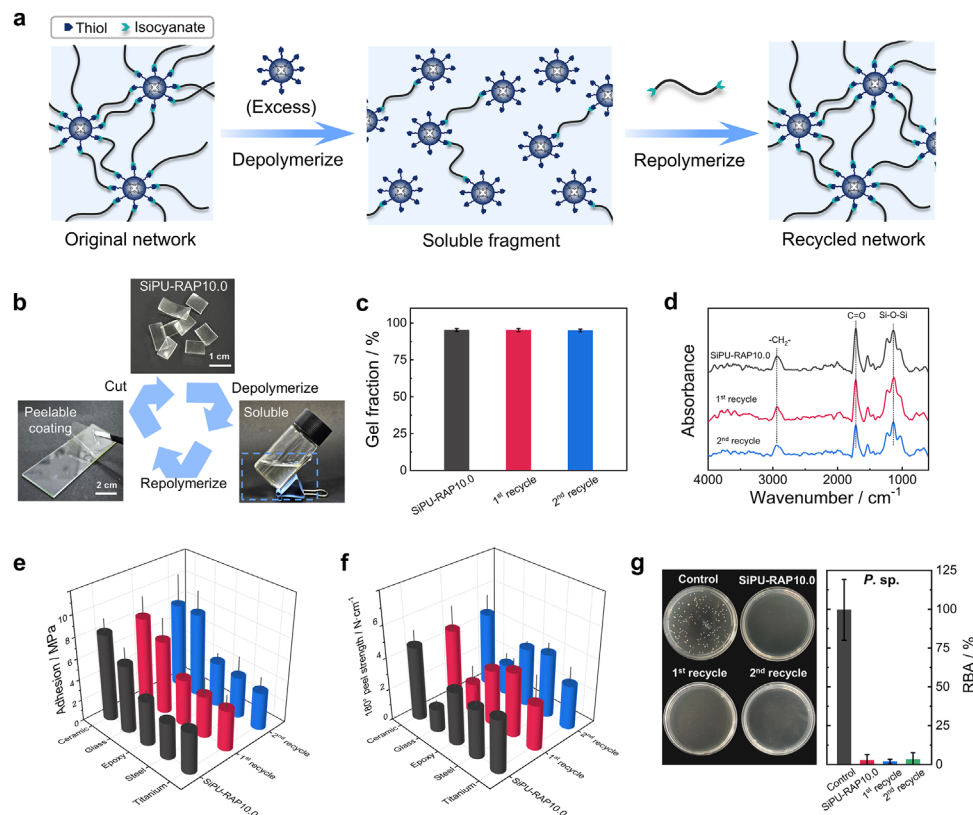


Figure 6. a) Reaction scheme of the recycling of SiPU-RAP_x coatings. b) Recycling process of the SiPU-RAP10.0 samples. c) Gel fraction of the original and recycled SiPU-RAP10.0 samples in the *N,N*-dimethylacetamide solvent. d) ATR-IR spectra of the original and recycled SiPU-RAP10.0 coatings. e) Adhesion strength and f) 180° peel strength of the original and recycled SiPU-RAP10.0 coatings on various substrates. g) Agar plate photographs of bacterial colonies of *P. sp.* after incubation with the original and recycled SiPU-RAP10.0 coatings and the corresponding RBA.

adhesion.^[41] Figure 4h shows that the 180° peel strength changed slightly with the RAP content, but SiPU-RAP10.0 still exhibited low peel strengths when attached to ceramic ($\approx 4.6 \text{ N}\cdot\text{cm}^{-1}$), glass ($\approx 1.5 \text{ N}\cdot\text{cm}^{-1}$), epoxy ($\approx 3.4 \text{ N}\cdot\text{cm}^{-1}$), steel ($\approx 3.1 \text{ N}\cdot\text{cm}^{-1}$), and titanium ($\approx 3.3 \text{ N}\cdot\text{cm}^{-1}$). The peel strength was lower on the glass substrate than on the other materials, which can be attributed to its smooth and flat surface. These results indicate that the synergistic modular construction and self-enrichment strategy effectively mitigated the adverse effects of excess antifouling additives on the mechanical properties of the coatings, thus ensuring high adhesion and on-demand peelability of gradient antifouling coatings.

The initial stage of biofouling typically involves protein adsorption, and fouling organisms secrete a protein-based bioadhesive to settle on a surface. Thus, the anti-protein performance of SiPU-RAP_x was evaluated by quartz crystal microbalance with dissipation (QCM-D) analysis. Figure 5a shows the frequency shift (Δf) and dissipation shift (ΔD) of the coatings overtime during the adsorption of fibrinogen, which has a high molecular weight ($M_w = 340 \text{ kDa}$) and adsorbs strongly on hydrophobic interfaces. For SiPU-RAP0 coating, a notable decrease in Δf and an increase in ΔD were observed with the introduction of the fibrinogen solution, indicating considerable fibrinogen adsorption on the surface. Increasing the RAP content decreased the magnitudes of Δf and ΔD . In particular, SiPU-RAP7.5 and SiPU-

RAP10.0 showed negligible Δf and ΔD , which indicates their superior protein resistance.

To verify the anti-biofouling properties of SiPU-RAP_x coatings, antibacterial assays, and anti-diatom tests were performed. As shown in Figure 5b, numerous viable colonies of gram-positive bacteria *Staphylococcus aureus* (*S. aureus*), gram-negative bacteria *Escherichia coli* (*E. coli*), and marine bacteria *Pseudomonas sp.* (*P. sp.*) were observed on the control and SiPU-RAP0 plates, which suggested that these coatings were highly susceptible to bacterial adhesion. Increasing the RAP content greatly reduced the number of bacterial colonies, and almost no bacteria ($<10\%$ adhesion) were detected on the SiPU-RAP7.5 and SiPU-RAP10.0 plates (Figure 5c–e). The anti-diatom (*N. incerta*) experiment showed a similar trend as the antibacterial experiment. A high density of diatoms was observed on the surface of the control and SiPU-RAP0 samples (Figure 5b). The number of diatoms adhering to the samples decreased as the RAP content increased, and the relative density of diatoms on both SiPU-RAP7.5 and SiPU-RAP10.0 samples was less than 10% (Figure 5f). The durable antifouling ability of SiPU-RAP_x coatings was further investigated through the anti-biofilm test after a certain period in artificial seawater (Figure S21, Supporting Information). For the evaluation, crystal violet is used to stain the biofilm on the samples, and the absorbance value is proportional to the amount of biofilm stained. On the control surface (polystyrene), the biofilm grew well with

an absorbance of ≈ 2.64 , whereas the biofilm on SiPU-RAP0 had an absorbance of ≈ 2.66 , showing poor resistance to biofilm. As the RAP content increased, significantly less biofilm accumulated on SiPU-RAP x . SiPU-RAP7.5 and SiPU-RAP10.0 exhibited excellent anti-biofilm ability with absorbance of ≈ 0.21 and ≈ 0.15 , respectively. These results indicate that the self-enriched RAP plays a key role in enhancing resistance to biofouling. Thus, increasing the RAP content was found to progressively enhance the antifouling properties of SiPU-RAP x .

3.1. Recyclability

Thiourethane bonds have been reported to undergo dissociative bond exchange at mild temperatures and stoichiometric conditions,^[42–44] which can be exploited to recycle cross-linked polythiourethane (Figure 6a). SiPU-RAP10.0 was peeled off a substrate and then cut into small pieces and mixed with excess POSS-SH in a solvent along with 1,5-diazabicyclo[4.3.0]non-5-ene (DBN) as a catalyst. Compared with basic and nucleophilic catalysts, the dual-capable basic and nucleophilic catalyst DBN demonstrates higher catalytic efficiency and facilitates the reversibility of the dynamic covalent polythiourethane network through the synergetic associative and dissociative thiourethane bond exchange mechanism.^[45,46] This mixture was stirred at 120 °C, and complete dissolution of the pieces was observed within 1 h. The base-catalyzed associative dynamic mechanism of the thiourethane bonds was followed by depolymerization in the presence of excess thiols. After adding a stoichiometric amount of NCO-PU and RAP to the thiol oligomer and recoating it onto the substrate surface, the recycled coatings were obtained (Figure 6b). Following two cycles of this process, the materials displayed comparable gel contents, highlighting the cross-linked nature of the recycled polymer network (Figure 6c). Moreover, the chemical structure of SiPU-RAP10.0 remained unchanged before and after multiple recycling cycles, as evidenced by attenuated total reflection infrared (ATR-IR) spectra (Figure 6d). Even after the second solvolytic recycle, SiPU-RAP10.0 coating maintained high substrate adhesion and easy peelability toward various substrates including ceramic, glass, epoxy, steel, and titanium (Figure 6e,f). The corresponding recycling efficiencies of the adhesion strength and 180° peel strength of SiPU-RAP10.0 were shown in Figures S22,S23 (Supporting Information). The values were all higher than 90%, indicating the good recyclability of the coating. Figure 6g shows that almost no bacteria (<10% bacterial adhesion) were observed on the recycled SiPU-RAP10.0, which indicates that the antifouling ability was retained. This was attributed to RAP self-enriching on the coating during the recycling process. It should be mentioned that this strategy well addresses the need for on-demand temporary or durable antifouling and end-of-life materials can be recycled through solvolytic recycling, thereby enhancing sustainability.

4. Conclusion

We developed a gradient antifouling coating with high adhesion, on-demand peelability, and superior recyclability by a novel modular construction strategy between POSS-SH (fundamental

module), NCO-PU (fundamental module), and RAP (functional module). The high adhesion and on-demand peelability of the coatings were realized by the self-reinforcing effect of POSS-SH and the strong hydrogen bond interaction of the polymer network. The dynamic covalent cross-linking between modules facilitated the recyclability of the coatings. Meanwhile, the incorporated RAP module could self-enrich on the surface of the coating to exhibit gradient antifouling ability. Compared with traditional antifouling coatings, we have solved the trade-off of the coexistence of high adhesion, on-demand peelability, recyclability, and surface antifouling functionality by the modular construction, utilization of self-reinforcing POSS, and self-enrichment strategy. Overall, this study offers a new perspective on the design and fabrication of peelable antifouling coatings that can be applied to on-demand temporary or durable antifouling of equipment in the marine and medical fields.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The financial support of the Key-Area Research and Development Program of Guangdong Province (2023B0101200006), the Basic and Applied Basic Research Foundation of Guangdong Province (2023B1515020025), the National Natural Science Foundation of China (52273073 and U2241286), and the National Key Research and Development Program of China (2022YFB3806403) is acknowledged.

Conflict of Interest

The authors declare that they have no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

antifouling, modular construction, peelable coating, recyclable, thiourethane

Received: April 3, 2025
Revised: May 8, 2025
Published online: June 1, 2025

- [1] H. C. Jin, L. M. Tian, W. Bing, J. Zhao, L. Q. Ren, *Prog. Mater. Sci.* **2022**, *124*, 100889.
- [2] W. L. Li, E. S. Thian, M. Wang, Z. Y. Wang, L. Ren, *Adv. Sci.* **2021**, *8*, 2100368.
- [3] H. Luo, X. Q. Yin, P. F. Tan, Z. P. Gu, Z. M. Liu, L. Tan, *J. Mater. Chem. B* **2021**, *9*, 2802.
- [4] B. N. Sahoo, P. J. Thomas, P. Thomas, M. M. Greve, *ACS Sens.* **2025**, *10*, 1600.

- [5] J. A. Callow, M. E. Callow, *Nat. Commun.* **2011**, 2, 244.
- [6] X. R. Zang, Y. M. Ni, Q. Y. Wang, Y. Cheng, J. Y. Huang, X. W. Cao, C. J. Carmalt, Y. K. Lai, D. Ha Kim, Y. J. Liu, Z. Q. Lin, *Mater. Today* **2024**, 75, 210.
- [7] R. Ciriminna, F. V. Bright, M. Pagliaro, *ACS Sustainable Chem. Eng.* **2015**, 3, 559.
- [8] M. S. Selim, S. A. El-Safty, M. A. Shenashen, S. A. Higazy, A. Elmarakbi, *J. Mater. Chem. B* **2020**, 8, 3701.
- [9] E. Guazzelli, F. Perondi, F. Criscitiello, C. Pretti, M. Oliva, V. Casu, F. Maniero, L. Gazzera, G. Galli, E. Martinelli, *J. Mater. Chem. B* **2020**, 8, 9764.
- [10] W. W. Zheng, J. Y. Huang, X. R. Zang, X. F. Xu, W. L. Cai, Z. Q. Lin, Y. K. Lai, *Adv. Mater.* **2022**, 34, 2204581.
- [11] C. F. Ma, H. Zhou, B. Wu, G. Z. Zhang, *ACS Appl. Mater. Interfaces* **2011**, 3, 455.
- [12] G. Galli, E. Martinelli, *Macromol. Rapid Commun.* **2017**, 38, 1600704.
- [13] J. S. Pan, X. Q. Ai, C. F. Ma, G. Z. Zhang, *Acc. Chem. Res.* **2022**, 55, 1586.
- [14] L. R. Chen, Y. Y. Duan, M. Cui, R. L. Huang, R. X. Su, W. Qi, Z. M. He, *Sci. Total Environ.* **2021**, 766, 144469.
- [15] X. R. Zang, X. W. Cao, W. W. Zheng, T. X. Zhu, Y. G. Lei, J. Y. Huang, Z. Chen, L. Teng, J. Bian, Y. K. Lai, *Chem. Eng. J.* **2023**, 451, 138573.
- [16] X. R. Zang, B. Jiang, Y. M. Ni, W. W. Zheng, T. X. Zhu, C. Zhong, X. W. Cao, J. Y. Huang, Y. Lai, Z. Q. Lin, *Adv. Sci.* **2024**, 11, 2305839.
- [17] P. Hu, Q. Y. Xie, C. F. Ma, G. Z. Zhang, *Langmuir* **2020**, 36, 2170.
- [18] M. Lejars, A. Margailan, C. Bressy, *Chem. Rev.* **2012**, 112, 4347.
- [19] E. Ralston, H. Gardner, K. Z. Hunsucker, G. Swain, *Front. Mar. Sci.* **2022**, 9, 836555.
- [20] D. Howell, B. Behrends, in *Biofouling*, John Wiley & Sons, Ltd., **2009** UK, 16.
- [21] J. S. Zhang, Q. Q. Shang, Y. Hu, G. Q. Zhu, J. Huang, X. X. Yu, J. W. Cheng, C. G. Liu, J. Q. Chen, G. D. Feng, Y. H. Zhou, *Prog. Org. Coat.* **2022**, 165, 106742.
- [22] P. C. Uzoma, P. Y. Wu, N. J. Gao, X. L. Ding, F. C. Liu, O. V. Penkov, H. Hu, *Adv. Eng. Mater.* **2025**, 27, 2402617.
- [23] J. C. Liu, J. Hu, J. Li, Y. Zhang, X. J. Li, *Prog. Org. Coat.* **2023**, 184, 107865.
- [24] P. G. Wagle, S. S. Tamboli, A. P. More, *Prog. Org. Coat.* **2021**, 150, 106005.
- [25] H. Y. Qiu, K. Feng, A. Gapeeva, K. Meurisch, S. Kaps, X. Li, L. M. Yu, Y. K. Mishra, R. Adelung, M. Baum, *Prog. Polym. Sci.* **2022**, 127, 101516.
- [26] J. P. Chen, R. K. Jian, K. Yang, W. B. Bai, C. M. Huang, Y. C. Lin, B. B. Zheng, F. F. Wei, Q. Lin, Y. L. Xu, *J. Clean Prod.* **2021**, 318, 128527.
- [27] Q. H. Zhang, H. He, K. Xi, X. Huang, X. H. Yu, X. D. Jia, *Macromolecules* **2011**, 44, 550.
- [28] M. H. Liu, F. H. Liu, X. B. Xu, D. F. Yu, I. Wyman, H. Yang, J. B. Wang, X. Wu, *ACS Appl. Mater. Interfaces* **2019**, 11, 16914.
- [29] R. Z. Chen, Y. S. Zhang, Q. Y. Xie, Z. X. Chen, C. F. Ma, G. Z. Zhang, *Adv. Funct. Mater.* **2021**, 31, 2011145.
- [30] X. Lin, C. Y. Li, L. X. Liang, Q. Y. Guo, Y. Zhang, S. R. Fu, Q. Zhang, F. Chen, D. Han, Q. Fu, *Nat. Commun.* **2024**, 15, 9680.
- [31] X. H. Wang, Y. X. Li, Y. H. Qian, H. Qi, J. Li, J. Q. Sun, *Adv. Mater.* **2018**, 30, 1803854.
- [32] G. M. Choi, J. Jin, D. Shin, Y. H. Kim, J. H. Ko, H. G. Im, J. Jang, D. Jang, B. S. Bae, *Adv. Mater.* **2017**, 29, 1700205.
- [33] S. S. Choi, A. S. Lee, S. S. Hwang, K. Y. Baek, *Macromolecules* **2015**, 48, 6063.
- [34] J. L. Zhu, J. Y. Zheng, Y. Hu, J. Wei, Y. C. Ma, H. L. Gou, H. Fan, *Prog. Org. Coat.* **2023**, 185, 107907.
- [35] Z. Q. Zhang, Y. J. Huang, Q. Y. Xie, G. J. Liu, C. F. Ma, G. Z. Zhang, *Prog. Polym. Sci.* **2024**, 154, 101840.
- [36] K. K. Zhang, S. S. Huang, J. D. Wang, G. J. Liu, *Angew. Chem., Int. Ed.* **2019**, 58, 12004.
- [37] X. Lin, M. X. Nie, H. Liu, D. L. Zhou, S. R. Fu, Q. Zhang, D. Han, Q. Fu, *Chem. Mater.* **2023**, 36, 575.
- [38] P. Hu, R. Xie, Q. Y. Xie, C. F. Ma, G. Z. Zhang, *Chem. Eng. J.* **2022**, 449, 137875.
- [39] M. Wang, Q. Y. Xie, J. S. Pan, C. F. Ma, G. Z. Zhang, *Chem. Eng. J.* **2023**, 464, 142769.
- [40] Z. Q. Zhang, Q. Y. Xie, G. L. Zhang, C. F. Ma, G. Z. Zhang, *ACS Appl. Polym. Mater.* **2023**, 5, 3524.
- [41] P. L. Zhang, G. L. Zhang, J. S. Pan, C. F. Ma, G. Z. Zhang, *ACS Appl. Mater. Interfaces* **2023**, 15, 5998.
- [42] X. Lopez De Pariza, O. Varela, S. O. Catt, T. E. Long, E. Blasco, H. Sardon, *Nat. Commun.* **2023**, 14, 5504.
- [43] S. Lee, J. Lyu, A. Jeong, M. Jung, J. H. Kang, H. Jung, Y. I. Park, J. E. Jeong, J. C. Kim, *ACS Sustainable Chem. Eng.* **2024**, 12, 12386.
- [44] Z. Q. Zhang, Q. Y. Xie, G. L. Zhang, C. F. Ma, G. Z. Zhang, *Adv. Funct. Mater.* **2024**, 34, 202408748.
- [45] Z. B. Wen, X. Han, B. D. Fairbanks, K. K. Yang, C. N. Bowman, *Polymer* **2020**, 202, 122715.
- [46] C. J. Fan, Z. B. Wen, Z. Y. Xu, Y. Xiao, D. Wu, K. K. Yang, Y. Z. Wang, *Macromolecules* **2020**, 53, 4284.